

MERN STACK INTERNSHIP – DAY 19 ASSIGNMENT REPORT

Company

Insta Dot Analytics

Submitted By

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FOR SOURCE CODE:

GIT HUB :- <https://github.com/chayan-soni/MERN-Internship>

DAY 19 – Admin Panel & Product Management

Objective

The objective of this task was to develop a complete Admin Dashboard for the MERN E-Commerce application. The dashboard provides administrators with tools to manage products, monitor platform statistics, and securely access administrative functionalities through JWT authentication.

Technologies Used

Frontend

- React.js
- React Router DOM
- Axios
- Bootstrap / Tailwind CSS
- React Toastify

Backend

- Node.js
- Express.js
- MongoDB

- Mongoose
- JWT Authentication

Development Tools

- VS Code
- Postman
- Git & GitHub

Task Implementation

1. Admin Dashboard Development

A dedicated Admin Dashboard was created to provide centralized management of the e-commerce platform. The dashboard includes multiple sections for product management and business analytics.

Features:

- Secure Admin Login
- Dashboard Navigation Panel
- Statistics Overview Cards
- Product Management Interface
- Search and Filtering Options
- Protected Admin Routes

2. Product Management Module

Implemented complete CRUD operations for product management.

Add Product

- Admin can add new products.
- Product details include:
 - Product Name
 - Description
 - Price
 - Category
 - Stock Quantity
 - Product Image URL

Update Product

- Existing products can be edited.
- Product information updates are reflected instantly in the database.

Delete Product

- Admin can remove products from the catalog.

- Confirmation handling implemented before deletion.

View Products

- Product list displayed in a tabular format.
- Dynamic data fetched from MongoDB using REST APIs.

3. Dashboard Statistics

Implemented APIs and frontend components to display real-time platform statistics.

Statistics Displayed

Total Users

Displays the total number of registered users.

Total Orders

Shows total orders placed on the platform.

Total Products

Displays total products available in the inventory.

Total Revenue

Calculates and displays total revenue generated from orders.

4. Search Functionality

Implemented search functionality for products.

Features:

- Search by Product Name.
- Instant filtering of product list.
- Case-insensitive matching.
- Improved product management efficiency.

5. Sorting Functionality

Added sorting options to organize products.

Sorting Options:

- Product Name (A-Z)
- Product Name (Z-A)
- Price (Low to High)
- Price (High to Low)

6. Pagination

Implemented pagination for better handling of large datasets.

Benefits:

- Improved performance.
- Reduced page load time.
- Better user experience.
- Easy navigation through products.

7. JWT Protected Admin Routes

Implemented secure authentication and authorization using JSON Web Tokens (JWT).

Security Features:

- Token-based Authentication
- Admin Route Protection
- Unauthorized Access Prevention
- Secure API Access
- Middleware Verification

Workflow:

1. Admin logs in.
2. JWT token generated.
3. Token stored on client side.
4. Protected routes validate token.
5. Authorized users gain dashboard access.

Backend APIs Developed

Product APIs

Add Product

POST /api/products

Get All Products

GET /api/products

Get Product By ID

GET /api/products/:id

Update Product

PUT /api/products/:id

Delete Product

DELETE /api/products/:id

Dashboard Statistics APIs

Total Users

GET /api/admin/users/count

Total Orders

GET /api/admin/orders/count

Total Products

GET /api/admin/products/count

Total Revenue

GET /api/admin/revenue

Authentication APIs

Admin Login

POST /api/auth/login

Verify JWT Token

GET /api/auth/verify

Testing & Validation

Backend Testing

- Tested APIs using Postman.
- Verified CRUD operations.
- Validated JWT authentication.
- Checked error handling scenarios.

Frontend Testing

- Verified dashboard rendering.
- Tested search functionality.
- Tested sorting options.
- Tested pagination controls.
- Verified protected routes.

Challenges Faced

1. Managing JWT authentication across protected routes.
2. Implementing efficient pagination for large datasets.
3. Synchronizing dashboard statistics with database records.
4. Handling product updates without page refresh.
5. Maintaining consistent state across admin components.

Learning Outcomes

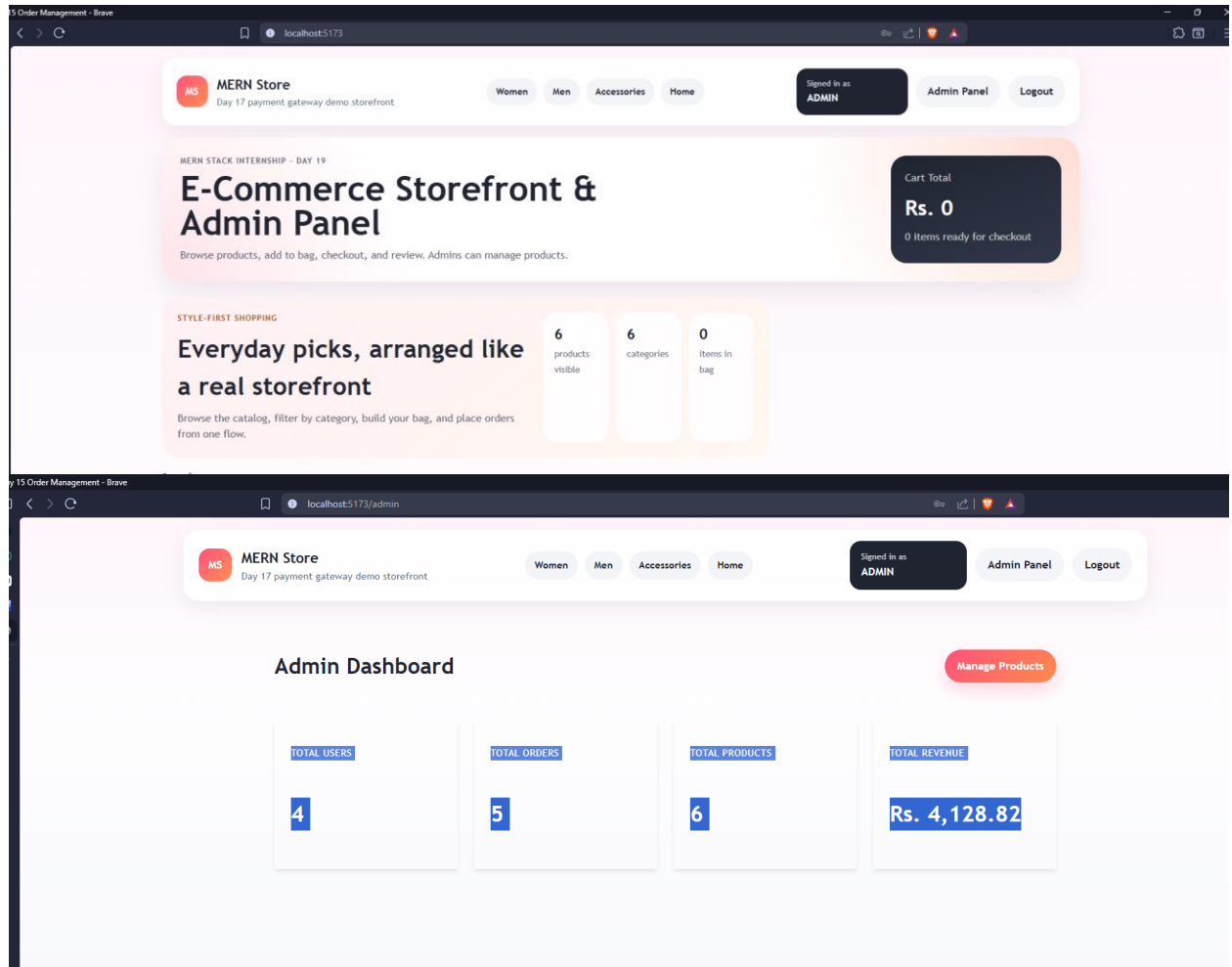
Through this task, I learned:

- Building Admin Dashboards in React.
- Implementing Product CRUD Operations.
- Creating Dashboard Analytics.
- JWT Authentication & Authorization.
- Search, Sorting, and Pagination Techniques.
- REST API Development with Express.js.
- MongoDB Data Management.
- Secure Route Protection in MERN Applications.

GitHub Repository

(Insert GitHub Repository Link Here)

Screenshots



Compass

{ } My Queries

⚙ Data Modeling

CONNECTIONS (1)

Search connections

▼ localhost:27017

▶ admin

▶ config

▼ day15_order_management

└─ orders

└─ payments

└─ products

└─ reviews

└─ **users** ...

▶ day3db

▶ hirespec

▼ jwt-auth

└─ users

▶ local

▶ mern-day6

▶ product-management

▶ wishlist-management

users

+

localhost:27017 > day15_order_management > users

Documents 4

Aggregations

Schema

Indexes 2

Validation

🕒 Type a query: { field: 'value' } or [Generate query](#) ⚡

➕ ADD DATA ▼

📄 BULK ▼

📄 EXPORT DATA ▼

📄 EXPORT CODE

```
_id: ObjectId('6a345d7209287dc84f032fe1')
name: "Chayan Soni"
email: "chayansoni04@gmail.com"
password: "$2a$10$0R2BSMFbLGRMtGTxrR8CEus53yPkjq1xJGThAygDtTrhmUKjzwS"
createdAt: 2026-06-18T21:04:50.816+00:00
updatedAt: 2026-06-18T21:04:50.816+00:00
__v: 0
```

```
_id: ObjectId('6a3470cc514cf39fd1b80a7b')
name: "Hello"
email: "hello@gmail.com"
password: "$2a$10$M/OhSaKNgyXcAvkz7fZyG.Lddqz.VW8.3VjzCdXvpLVmHDRqY30la"
createdAt: 2026-06-18T22:27:24.680+00:00
updatedAt: 2026-06-18T22:27:24.680+00:00
__v: 0
```

```
▶ _id: ObjectId('6a34e8f5f0b1586cee53e88f')
name: "nikhil"
email: "nikhil@gmail.com"
password: "$2a$10$vULxs6ssdyCGV.8pRcUhehoDBDyk6NUZLv11YSwcvZeZMMaW4u6"
createdAt: 2026-06-19T07:00:05.474+00:00
updatedAt: 2026-06-19T07:00:05.474+00:00
__v: 0
```

```
_id: ObjectId('6a3944c20d2140d39fade55a')
name: "ADMIN"
email: "admin@gmail.com"
password: "$2a$10$Tvn.dx/bb.vx2mTImv9pyu4hP3GMJ92ZTsaUbjScbPAAMMtK008VS"
isAdmin: true
createdAt: 2026-06-22T14:20:50.885+00:00
updatedAt: 2026-06-22T14:20:50.885+00:00
__v: 0
```


Order Management - Brave

localhost:5173/admin/products

MS

MERN Store

Day 17 payment gateway demo storefront

Women

Men

Accessories

Home

Signed in as

ADMIN

Admin Panel

Logout

Product Management

Back to Dashboard

+ Add Product

Search products...

Latest

ID	Name	Price	Category	Stock	Actions
6a346071...	Ceramic Coffee Mug Set	Rs. 1199	Home	25	Edit Delete
6a346071...	Ergonomic Office Chair	Rs. 8999	Furniture	0	Edit Delete
6a346071...	Keyboard	Rs. 3499	Accessories	17	Edit Delete
6a346071...	Smart Fitness Band	Rs. 2499	Wearables	16	Edit Delete
6a346071...	Portable SSD 1TB	Rs. 6999	Storage	14	Edit Delete
6a346071...	Noise Cancelling Headphones	Rs. 4999	Electronics	12	Edit Delete

Previous

Page 1 of 1

Next

key

Latest

ID	Name	Price	Category	Stock	Actions
6a346071...	Keyboard	Rs. 3499	Accessories	17	Edit Delete

Previous

Page 1 of 1

Next